Composing layers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable layer-composition record.

Trace hidden values,repair dimensions and explain information lost before the output.

Keep one habit: W1–W3/W5 → selected new network → project layer contract → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Dot products and weighted sums. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- A network connects **layer**s so that the **output** vector of one **layer** becomes the **input** vector of the next. Use a fictional numerical **input** x = [x1, x2].

  The first **layer** computes z1 = x1 - x2 and z2 = x1 + x2 - 2, then applies ReLU separately: h1 = max(0, z1), h2 = max(0, z2). Its **output** is the ordered vector h = [h1, h2].

- The next **layer** computes y = 2h1 - h2 + 1. It receives h, not the original x. It uses weights [2, -1] and bias 1, with identity as its **output** activation.

  This is a two-**layer** computation when counting the **layer**s with weights; an **input** placeholder is not counted here. All parameters are supplied, so this lesson performs no **training**.

- The two first-**layer** units both receive the same original x. They can be calculated in either order. Only after both activations are available do we evaluate the **output layer**.

Activation functions

> **My remembered idea, example and correction**

Dot products and weighted sums

> **My remembered idea, example and correction**

Lesson reference for workbook practice

Stage	Input size	Output size
First layer with ReLU	2 components	2 components
Output layer with identity	2 components	1 component

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main network x[x1, x2]:z1=x1−x2, z2=x1+x2−2; h=[ReLU(z1), ReLU(z2)]; y=2h1−h2+1 with identity output. Two weighted layers:2→2→1. Main x[3, 1]z[2, 2], h[2, 2], y3; [1, 3]z[−2, 2], h[0, 2], y−1; [1, 1]zeros, y1. For[3, 0], h[3, 1]correcty 6, swapped-hiddeny 0; [4, 1]h[3, 3]y4 cannot expose swap. Missing/invalid fields need earlier validation. No reference labels, so no accuracy claim.

  Separate pictured h1ReLU(u−2), h2ReLU(v−1), y=h1−h2:input[4, 3]y0; v→1 givesy2. Independent model has its own formulas.

- For x = [1, 3], write both scores, both ReLU activations and y using the lesson model.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main network x[x1, x2]:z1=x1−x2, z2=x1+x2−2; h=[ReLU(z1), ReLU(z2)]; y=2h1−h2+1 with identity output. Two weighted layers:2→2→1. Main x[3, 1]z[2, 2], h[2, 2], y3; [1, 3]z[−2, 2], h[0, 2], y−1; [1, 1]zeros, y1. For[3, 0], h[3, 1]correcty 6, swapped-hiddeny 0; [4, 1]h[3, 3]y4 cannot expose swap. Missing/invalid fields need earlier validation. No reference labels, so no accuracy claim.

  Separate pictured h1ReLU(u−2), h2ReLU(v−1), y=h1−h2:input[4, 3]y0; v→1 givesy2. Independent model has its own formulas.

- A revised hidden **layer** returns [h1, h2, h3]. The **output layer** still has weights [2, -1]. Explain the mismatch and what must be specified before calculation.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main network x[x1, x2]:z1=x1−x2, z2=x1+x2−2; h=[ReLU(z1), ReLU(z2)]; y=2h1−h2+1 with identity output. Two weighted layers:2→2→1. Main x[3, 1]z[2, 2], h[2, 2], y3; [1, 3]z[−2, 2], h[0, 2], y−1; [1, 1]zeros, y1. For[3, 0], h[3, 1]correcty 6, swapped-hiddeny 0; [4, 1]h[3, 3]y4 cannot expose swap. Missing/invalid fields need earlier validation. No reference labels, so no accuracy claim.

  Separate pictured h1ReLU(u−2), h2ReLU(v−1), y=h1−h2:input[4, 3]y0; v→1 givesy2. Independent model has its own formulas.

- Compare hidden vectors for x = [4, 1] and x = [3, 0]. Which **input** reveals a bug that swaps h1 and h2 but keeps **output** weights unchanged? Give correct and swapped **output**s for that **input**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main network x[x1, x2]:z1=x1−x2, z2=x1+x2−2; h=[ReLU(z1), ReLU(z2)]; y=2h1−h2+1 with identity output. Two weighted layers:2→2→1. Main x[3, 1]z[2, 2], h[2, 2], y3; [1, 3]z[−2, 2], h[0, 2], y−1; [1, 1]zeros, y1. For[3, 0], h[3, 1]correcty 6, swapped-hiddeny 0; [4, 1]h[3, 3]y4 cannot expose swap. Missing/invalid fields need earlier validation. No reference labels, so no accuracy claim.

  Separate pictured h1ReLU(u−2), h2ReLU(v−1), y=h1−h2:input[4, 3]y0; v→1 givesy2. Independent model has its own formulas.

- Use the lesson model to verify h and y for [0, 1] and [0, 2]. Could a fixed later **layer** receiving only h distinguish the two original **input**s?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A new model takes [u, v]. Compute a = ReLU(u + v - 3), b = ReLU(u - v), then q = a + 2b - 1. Trace both **input**s [2, 1] and [1, 2], including scores before ReLU. Apply class 1 when q >= 0.

  State the hidden and **output** vector sizes.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Compose a new network independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled composing **layer**s **evidence** sheet to Image-pattern Network Investigation. Show how you can trace how one **layer** supplies the next **layer input**s.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Repair a dimension mismatch for Composing **layer**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Composing **layer**s. Proposed explanation: "The second hidden unit receives the first hidden unit **output**". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A new model takes [u, v]. Compute a = ReLU(u + v - 3), b = ReLU(u - v), then q = a + 2b - 1. Trace both **input**s [2, 1] and [1, 2], including scores before ReLU. Apply class 1 when q >= 0.

  State the hidden and **output** vector sizes.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

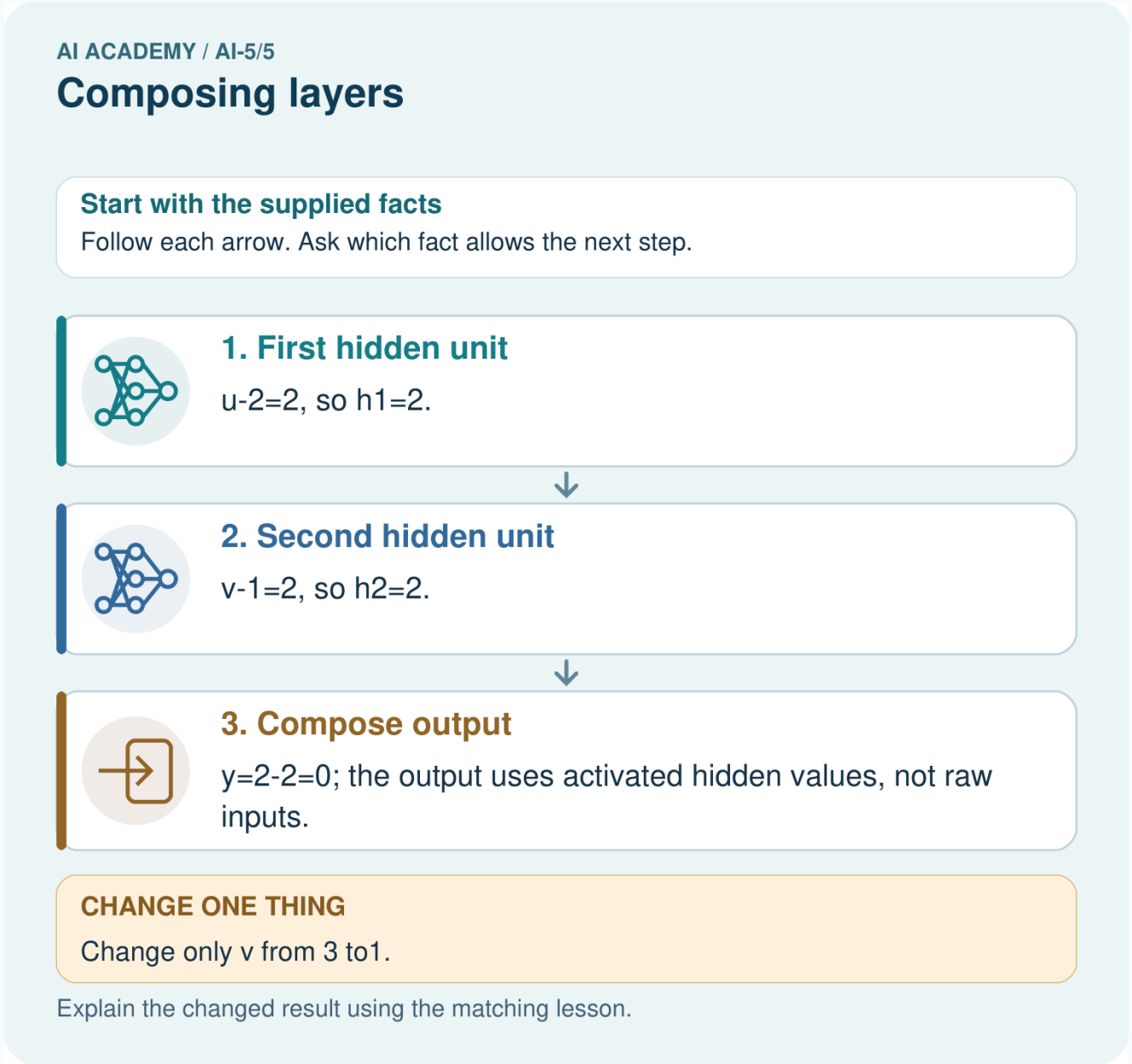
> Keep this task's inputs,rule and response together.

Complete task and evidence

- Compose two hidden detectors before one **output** instead of skipping a **layer**.
- For **input** [u, v], h1=ReLU(u-2), h2=ReLU(v-1), y=h1-h2. **Input** [4, 3].
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- What intermediate value explains the changed **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Compose two hidden detectors before one **output** instead of skipping a **layer**.
- Which labelled step matches this detail? y=2-2=0; the **output** uses activated hidden values, not raw **input**s.
- 1. Compose **output**
- 2. Second hidden unit

- 3. First hidden unit

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check after the first attempt

After your attempt, compare with the supported explanation: Compose **output** y=2-2=0; the **output** uses activated hidden values, not raw **input**s. **Composition** means one **layer**'s result becomes the next **layer**'s **input**.

Compute both hidden units before combining them. At [4,3], their equal **output**s cancel. Lowering only the second **input** removes its positive hidden contribution, so subtracting it no longer reduces the **output**.

The final subtraction must use activated hidden values rather than the raw measurements.

Changed-case question and optional hint

- Changed-case practice: Change only v from 3 to1. Which conclusion is supported by the supplied facts?

- 1. h1 stays 2, h2 becomes 0, y becomes 2.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: h1 stays 2, h2 becomes 0, y becomes 2. The second hidden unit's preactivation reaches 0 while the first unit is unchanged.

Repair and retain the record

Review Activation functions (ai-5/4). Goal: Trace identity ReLU and sigmoid on a numerical score. Short practice: Which original **input**s cannot be distinguished by this network's final **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dot products and weighted sums (ai-5/2). Goal: Calculate a dot product by matching feature positions. Short practice: Which connection explains why more right brightness lowers this score?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Composing layers: the case

AI ACADEMY / AI-5/5 · 01 · LOOK AT THE FACTS

Composing layers: the case

The situation

Compose two hidden detectors before one output instead of skipping a layer.

Supplied evidence

For input [u, v], h1=ReLU(u-2), h2=ReLU(v-1), y=h1-h2. Input [4, 3].
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain composing layers using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

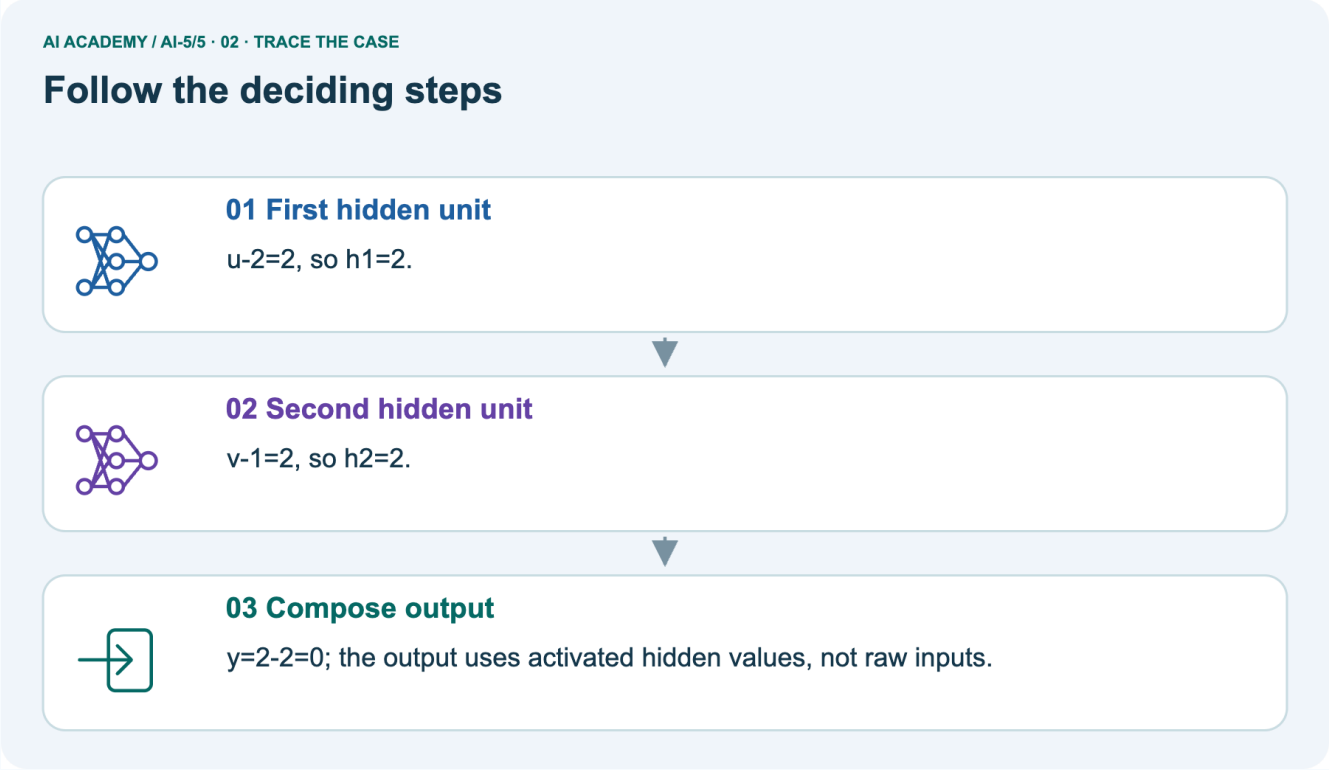
Compose two hidden detectors before one **output** instead of skipping a **layer**. Supplied facts: For **input** [u, v], h1=ReLU(u-2), h2=ReLU(v-1), y=h1-h2. **Input** [4, 3].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

First hidden unit: u-2=2, so h1=2. → Second hidden unit: v-1=2, so h2=2. → Compose **output**: y=2-2=0; the **output** uses activated hidden values, not raw **input**s.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Complete each stage before using its output

AI ACADEMY / AI-5/5 · 03 · INSPECT THE LESSON MODEL

Complete each stage before using its output

Available input	Computation	Value passed on
x is [3, 1]	Both first layer scores	z is [2, 2]
z is [2, 2]	ReLU on each component	h is [2, 2]
h is [2, 2]	2h1 - h2 + 1	y is 3

Read the rows from top to bottom. The first two rows together form the hidden layer; the final row is the output layer. Each row shows the values actually passed to the next stage.

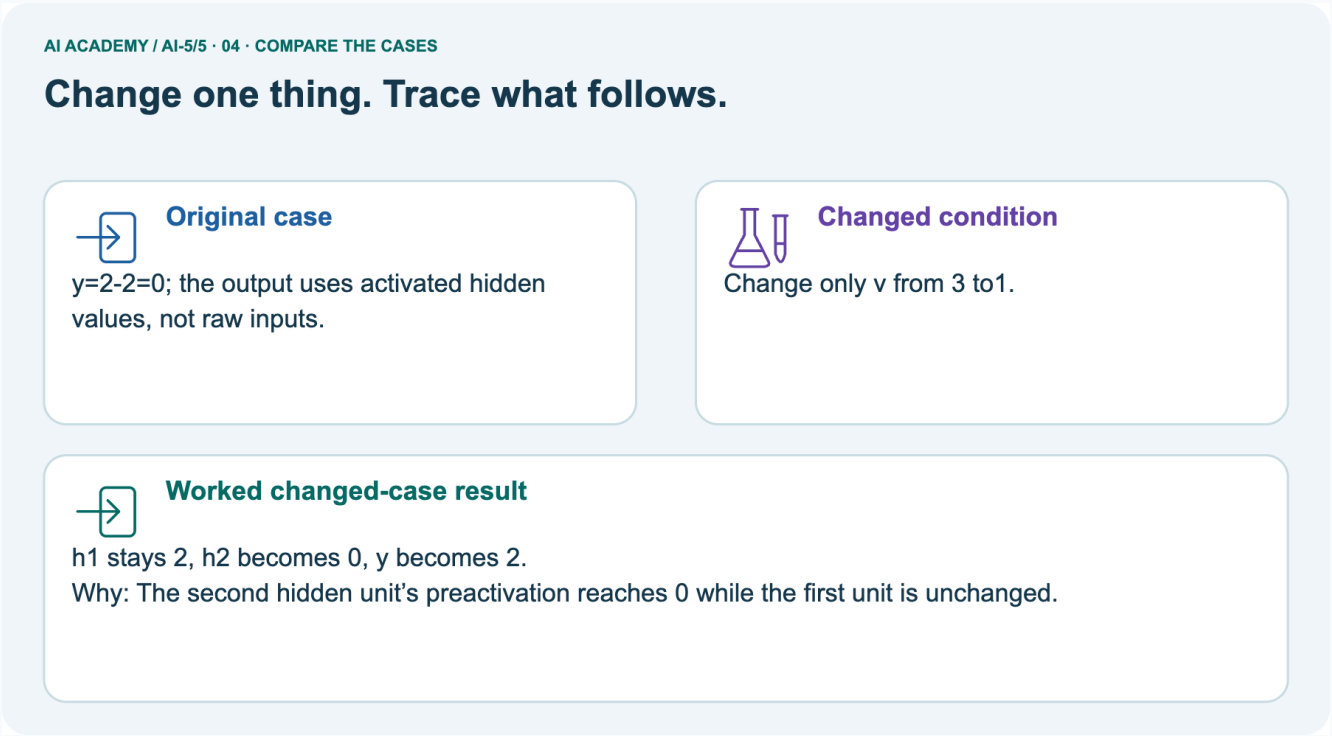
Follow the labelled picture with the matching explanation. Open the vector diagram

Complete each stage before using its **output**. x is [3, 1] | Both first **layer** scores | z is [2, 2]. z is [2, 2] | ReLU on each component | h is [2, 2]. h is [2, 2] | 2h1 - h2 + 1 | y is 3

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: y=2-2=0; the **output** uses activated hidden values, not raw **input**s. Changed condition: Change only v from 3 to1. Expected result: h1 stays 2, h2 becomes 0, y becomes 2.

Reason: The second hidden unit's preactivation reaches 0 while the first unit is unchanged.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit

AI ACADEMY / AI-5/5 · 05 · EXPLAIN WITH CARE

Connect the explanation to its limit

What the case explains

Composition means one layer's result becomes the next layer's input. Compute both hidden units before combining them. At [4,3], their equal outputs cancel. Lowering only the second input removes its positive hidden contribution, so subtracting it no longer reduces the output. The final subtraction must use activated hidden values rather than the raw measurements.

What it does not establish

Hand-composed detectors illustrate layer order; they are not evidence of a learned or biologically realistic neural mechanism.

Follow the labelled picture with the matching explanation. Open the vector diagram

Composition means one **layer**'s result becomes the next **layer**'s **input**. Compute both hidden units before combining them. At [4,3], their equal **output**s cancel.

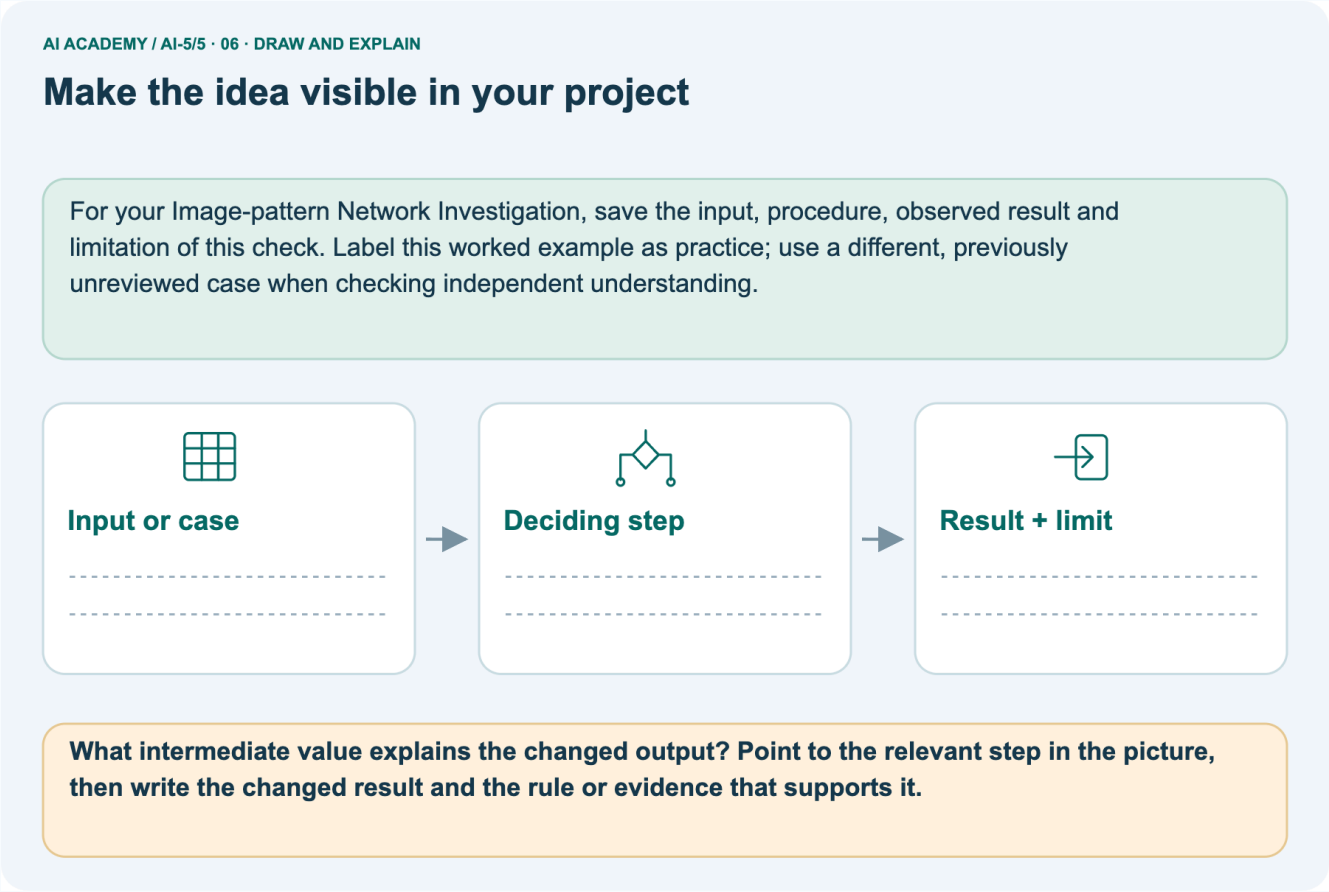
Lowering only the second **input** removes its positive hidden contribution, so subtracting it no longer reduces the **output**. The final subtraction must use activated hidden values rather than the raw measurements.

Limit: Hand-composed detectors illustrate **layer** order; they are not **evidence** of a learned or biologically realistic neural **mechanism**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What intermediate value explains the changed **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Earlier wording, alternatives and picture transcripts

01. Composing layers: the case

02. Follow the deciding steps

03. Complete each stage before using its output

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.